

Custom Claude Integration for Google Workspace — Google Sheets MCP Saves Hours Going Back and Forth

How to move from constant CSV downloads and copy-paste friction to Claude reading, writing, and creating Google Sheets autonomously

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5 steps

eliminated from every data task

0 €

ongoing cost

All sheets

accessible — not just one

01 · THE REAL PROBLEM — DEATH BY DATA FRICTION

THE GOAL

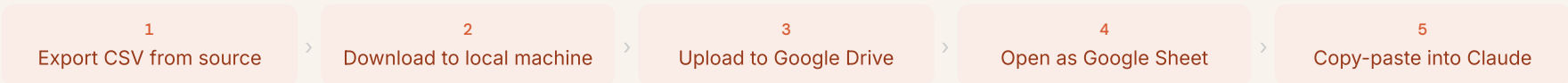
Deep integration of Claude across all business workflows. Not a one-off connection — a flexible, reusable bridge between Claude and **any Google Sheet**, for any task: grants database, social media analysis, webinar tracking, client data.

THE DAILY REALITY

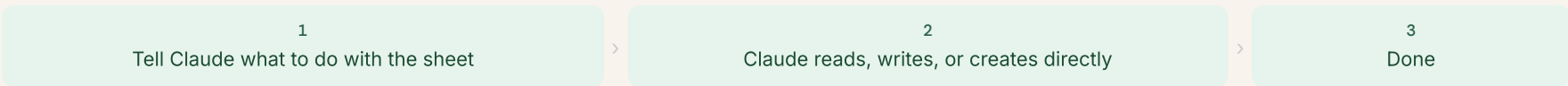
Every time data needed to move between Claude and a spreadsheet, it required a **full manual loop** — downloading, converting, uploading, sharing, then copying results back out. Slow, error-prone, and capacity-draining.

02 · BEFORE VS AFTER — THE WORKFLOW

Before



After



02B · THE HIDDEN MISCONCEPTION

COMMON ASSUMPTION THAT'S WRONG

Claude.ai has a native Google Drive connector — so surely it can access Google Sheets? **No**. The native Drive connector is limited to Google Docs only. Claude cannot read cell contents, write to cells, or create new spreadsheets through it. For any real Sheets integration, a dedicated MCP server is required — in Claude Desktop, configured separately from claude.ai entirely.

03 · BLOCKERS AND SOLUTIONS

BLOCKER

SOLUTION

✗ Service account key creation blocked — Google Workspace org policy had JSON key downloads disabled. Every standard tutorial assumes this works.	✓ Switched to OAuth 2.0 Client ID (Desktop app type) — bypasses org policy entirely, uses personal auth instead of service account keys.
✗ Google Drive API not enabled — mcp-google-sheets requires both Sheets API and Drive API. Enabling only one causes silent failures with no useful error message.	✓ Explicitly enabled both APIs in GCP Library — Sheets API and Drive API. They don't auto-enable each other.
✗ uvx not resolving in Claude Desktop config — using "uvx" as the command caused silent launch failures with no error output to debug from.	✓ Replaced with the full absolute path to uvx (e.g. /Users/name/.local/bin/uvx). Resolved immediately.

04 · WHAT CLAUDE CAN NOW DO WITH GOOGLE SHEETS

✓ Read cell contents directly — no upload, no copy-paste	✓ Write and update cells without manually touching the sheet
✓ Create brand new spreadsheets from scratch on command	✓ Works across all sheets — grants database, social media analysis, webinar tracking, and more

05 · FINAL SETUP — WHAT WAS BUILT

ARCHITECTURE

1

Google Cloud project — Sheets API + Drive API both enabled explicitly

2

OAuth 2.0 Client ID (Desktop app) — consent screen configured, personal email as test user

3

mcp-google-sheets added to claude_desktop_config.json — full absolute uvx path, OAuth credentials in env block

4

One-time browser auth flow — token saved automatically, no repeat login needed

06 · TECH STACK

Claude Desktopmcp-google-sheetsGoogle Sheets APIGoogle Drive APIOAuth 2.0uvxGoogle Workspace Business Standard

Native Drive connector — Docs only, not Sheets

07 · WHAT I'D DO DIFFERENTLY

HONEST REFLECTION

Start with OAuth from day one — even without org policy restrictions, it's more portable and avoids the security risk of stored JSON key files. Treat Claude Desktop and claude.ai as completely separate systems from the start — different configs, different MCP connections, nothing shared. Always use absolute paths in Claude Desktop configs; "uvx" alone is a silent failure waiting to happen.

Full step-by-step walkthrough published at thesciencetalk.com · More case studies at pranoti.thesciencetalk.com

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